Experiment 011: V2 Cells Before Q

Jordan Quillen Homology Project

Scope

Experiment ID: `EXP-011-v2-cells-no-Q`.

Target algebra: $Jord < x, y > /(x^2, xy, y^2)$.

The computation uses $A^{(1)} = \mathrm{Jord}\langle x, y, r_{xx}, r_{xy}, r_{yy}\rangle$ with $|x| = |y| = 0$, $|r_{xx}| = |r_{xy}| = |r_{yy}| = 1$, and $d(r_{xx}) = x^2$, $d(r_{xy}) = xy$, $d(r_{yy}) = y^2$.

The functor Q is not applied in this experiment.

The layers $V_3, V_4, \ldots$ are not constructed.

Backend convention: this run follows the existing ordinary commutative nonassociative low-weight Jordan backend with a signed derivation differential. It is backend-relative bounded evidence, not a fully graded operadic convention audit.

Finite-Weight Boundary

All conclusions in this report are restricted to weights with status completed. Weights not tested, skipped, or failed are not used to infer any global statement. This computation does not prove stabilization, termination, or absence of further degree 2 attaching generators in untested weights.

Weight Summary

w	$\dim C_0$	$\dim C_1$	$\dim C_2^{\mathrm{old}}$	$\mathrm{rank}\, d_1$	$\mathrm{rank}\, d_2$	$\dim Z_1$	$\dim H_1^{\mathrm{old}}$	status
1	2	0	0	0	0	0	0	completed
2	3	3	0	3	0	0	0	completed
3	6	6	0	6	0	0	0	completed
4	10	21	6	10	3	11	8	completed
5	24	48	30	24	24	24	0	completed
6	72	159	111	72	87	87	0	completed
7	220	552	444	220	326	332	6	completed
8	762	2064	1926	762	1300	1302	2	completed
9	–	–	–	–	–	–	–	skipped
10	–	–	–	–	–	–	–	skipped
11	–	–	–	–	–	–	–	skipped
12	–	–	–	–	–	–	–	skipped
13	–	–	–	–	–	–	–	skipped
14	–	–	–	–	–	–	–	skipped
15	–	–	–	–	–	–	–	skipped
16	–	–	–	–	–	–	–	skipped
17	–	–	–	–	–	–	–	skipped
18	–	–	–	–	–	–	–	skipped
19	–	–	–	–	–	–	–	skipped
20	–	–	–	–	–	–	–	skipped

Skipped, Failed, And Not Tested Weights

```
skipped_weights = [9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
failed_weights = []
not_tested_weights_in_requested_range = []
```

Formal Degree 2 Cell Candidates

Each listed record is a formal candidate only: the cell is not attached during this experiment.

s2_00001_w4

```
degree = 2
weight = 4
d(s2_00001_w4) = 1/2*((r_xx*x)*y) - 3/2*((r_xx*y)*x) + ((r_xy*x)*x)
boundary_remainder_mod_B_old = 1/2*((r_xx*x)*y) - 3/2*((r_xx*y)*x) + ((r_xy*x)*x)
selection_remainder_mod_B_old_plus_accepted_reps = 1/2*((r_xx*x)*y) - 3/2*((r_xx*y)*x) + ((r_xy*x)*x)
```

s2_00002_w4

```
degree = 2
weight = 4
d(s2_00002_w4) = -((r_xx*y)*y) + 2*((r_xy*x)*y) - 2*((r_xy*y)*x) + ((r_yy*x)*x)
boundary_remainder_mod_B_old = -((r_xx*y)*y) + 2*((r_xy*x)*y) - 2*((r_xy*y)*x) + ((r_yy*x)*x)
selection_remainder_mod_B_old_plus_accepted_reps = -((r_xx*y)*y) + 2*((r_xy*x)*y) - 2*((r_xy*y)*x) +
↪  ((r_yy*x)*x)
```

s2_00003_w4

```
degree = 2
weight = 4
d(s2_00003_w4) = 2*((r_xy*y)*y) - 3*((r_yy*x)*y) + ((r_yy*y)*x)
boundary_remainder_mod_B_old = 2*((r_xy*y)*y) - 3*((r_yy*x)*y) + ((r_yy*y)*x)
selection_remainder_mod_B_old_plus_accepted_reps = 2*((r_xy*y)*y) - 3*((r_yy*x)*y) + ((r_yy*y)*x)
```

s2_00004_w4

```
degree = 2
weight = 4
d(s2_00004_w4) = -((r_xx*x)*x) + ((x*x)*r_xx)
boundary_remainder_mod_B_old = -((r_xx*x)*x) + ((x*x)*r_xx)
selection_remainder_mod_B_old_plus_accepted_reps = -((r_xx*x)*x) + ((x*x)*r_xx)
```

s2_00005_w4

```
degree = 2
weight = 4
d(s2_00005_w4) = -((r_xx*y)*x) + ((x*x)*r_xy)
boundary_remainder_mod_B_old = -((r_xx*y)*x) + ((x*y)*r_xx)
selection_remainder_mod_B_old_plus_accepted_reps = -((r_xx*y)*x) + ((x*y)*r_xx)
```

s2_00006_w4

```
degree = 2
weight = 4
d(s2_00006_w4) = -1/2*((r_xx*y)*y) - ((r_xy*y)*x) + 1/2*((x*x)*r_yy) + ((x*y)*r_xy)
boundary_remainder_mod_B_old = -1/2*((r_xx*y)*y) - ((r_xy*y)*x) + ((x*y)*r_xy) + 1/2*((y*y)*r_xx)
selection_remainder_mod_B_old_plus_accepted_reps = -((r_xy*x)*y) - 1/2*((r_yy*x)*x) + ((x*y)*r_xy) +
↪  1/2*((y*y)*r_xx)
```

s2_00007_w4

```
degree = 2
weight = 4
d(s2_00007_w4) = -((r_yy*x)*y) + ((x*y)*r_yy)
boundary_remainder_mod_B_old = -((r_yy*x)*y) + ((y*y)*r_xy)
selection_remainder_mod_B_old_plus_accepted_reps = -((r_yy*x)*y) + ((y*y)*r_xy)
```

s2_00008_w4

```
degree = 2
weight = 4
d(s2_00008_w4) = -((r_yy*y)*y) + ((y*y)*r_yy)
boundary_remainder_mod_B_old = -((r_yy*y)*y) + ((y*y)*r_yy)
selection_remainder_mod_B_old_plus_accepted_reps = -((r_yy*y)*y) + ((y*y)*r_yy)
```

s2_00009_w7

```
degree = 2
weight = 7
d(s2_00009_w7) = -(((((x*x)*x)*x)*x)*r_xx) + ((((x*x)*x)*x)*(r_xx*x))
boundary_remainder_mod_B_old = -1/2*(((r_xx*x)*(x*x))*(x*x)) + 1/2*(((x*x)*r_xx)*((x*x)*x))
selection_remainder_mod_B_old_plus_accepted_reps = -1/2*(((r_xx*x)*(x*x))*(x*x)) +
↪  1/2*(((x*x)*r_xx)*((x*x)*x))
```

s2_00010_w7

```
degree = 2
weight = 7
d(s2_00010_w7) = 4*(((((x*x)*x)*x)*x)*r_xy) + 2*(((((x*x)*x)*x)*y)*r_xx) + 2*(((((x*x)*x)*y)*r_xx)*x) +
↪  4*(((((x*y)*x)*x)*x)*r_xx) - ((((x*x)*r_xx)*y)*(x*x)) + 2*((((x*x)*x)*r_xx)*(x*y)) -
↪  2*((((x*x)*x)*r_xy)*(x*x)) - 2*((((x*x)*x)*x)*(r_xx*y)) - 8*((((x*x)*x)*x)*(r_xy*x)) -
↪  2*((((x*x)*x)*y)*(r_xx*x)) + ((((x*x)*y)*(x*x))*r_xx)
boundary_remainder_mod_B_old = 12*(((r_xx*x)*(x*y))*(x*x)) + 12*(((r_xx*x)*x)*((x*y)*x)) +
↪  3*(((r_xx*y)*(x*x))*(x*x)) + 6*(((r_xy*x)*(x*x))*(x*x)) - 12*(((r_xy*x)*x)*((x*x)*x)) -
↪  3*(((x*x)*r_xx)*((x*x)*y)) - 6*(((x*x)*r_xx)*((x*y)*x)) - 12*(((x*x)*x)*((x*y)*r_xx))
selection_remainder_mod_B_old_plus_accepted_reps = 12*(((r_xx*x)*(x*y))*(x*x)) + 12*(((r_xx*x)*x)*((x*y)*x)) +
↪  3*(((r_xx*y)*(x*x))*(x*x)) + 6*(((r_xy*x)*(x*x))*(x*x)) - 12*(((r_xy*x)*x)*((x*x)*x)) -
↪  3*(((x*x)*r_xx)*((x*x)*y)) - 6*(((x*x)*r_xx)*((x*y)*x)) - 12*(((x*x)*x)*((x*y)*r_xx))
```

s2_00011_w7

```
degree = 2
weight = 7
d(s2_00011_w7) = (((((x*y)*x)*x)*x)*r_yy) + 2*(((((x*y)*x)*x)*y)*r_xy) + 4*(((((x*y)*x)*y)*r_xy)*x) -
↪  2*(((((x*y)*x)*y)*x)*r_xy) - (((((x*y)*x)*y)*y)*r_xx) - ((((x*x)*r_xy)*y)*(x*y)) +
↪  1/2*((((x*y)*x)*r_xx)*(y*y)) - 1/2*((((x*y)*x)*r_yy)*(x*x)) - 2*((((x*y)*x)*x)*(r_xy*y)) -
↪  ((((x*y)*x)*x)*(r_yy*x)) + ((((x*y)*x)*y)*(r_xx*y)) - 2*((((x*y)*x)*y)*(r_xy*x)) +
↪  ((((x*y)*y)*(x*x))*r_xy)
boundary_remainder_mod_B_old = (((r_xx*x)*(y*y))*(x*y)) - (((r_xx*x)*y)*((x*y)*y)) +
↪  1/4*(((r_xx*y)*(x*x))*(y*y)) + (((r_xx*y)*(x*y))*(x*y)) + 1/4*(((r_xx*y)*(y*y))*(x*x)) -
↪  (((r_xx*y)*x)*((x*y)*y)) - (((r_xx*y)*y)*((x*y)*x)) + 2*(((r_xy*x)*(x*y))*(x*y)) +
↪  (((r_xy*x)*(y*y))*(x*x)) + 2*(((r_xy*x)*x)*((y*y)*x)) + (((r_xy*x)*y)*((x*x)*y)) +
↪  2*(((r_xy*y)*(x*x))*(x*y)) + (((r_xy*y)*x)*((x*x)*y)) + (((r_xy*y)*y)*((x*x)*x)) +
↪  3/2*(((r_yy*x)*(x*x))*(x*y)) - 1/2*(((r_yy*x)*(x*y))*(x*x)) - 2*(((r_yy*x)*x)*((x*y)*x)) +
↪  1/4*(((r_yy*y)*(x*x))*(x*x)) - 1/4*(((x*x)*r_xx)*((y*y)*y)) - 1/2*(((x*x)*y)*((y*y)*r_xx)) -
↪  (((x*y)*(x*y))*(r_xx*y)) - (((x*y)*(y*y))*(r_xx*x)) - 2*(((x*y)*r_xx)*((x*y)*y)) -
↪  (((x*y)*r_xx)*((y*y)*x)) - 2*(((x*y)*r_xy)*((x*y)*x)) - (((x*y)*x)*((y*y)*r_xx))
selection_remainder_mod_B_old_plus_accepted_reps = (((r_xx*x)*(y*y))*(x*y)) - (((r_xx*x)*y)*((x*y)*y)) +
↪  1/4*(((r_xx*y)*(x*x))*(y*y)) + (((r_xx*y)*(x*y))*(x*y)) + 1/4*(((r_xx*y)*(y*y))*(x*x)) -
↪  (((r_xx*y)*x)*((x*y)*y)) - (((r_xx*y)*y)*((x*y)*x)) + 2*(((r_xy*x)*(x*y))*(x*y)) +
↪  (((r_xy*x)*(y*y))*(x*x)) + 2*(((r_xy*x)*x)*((y*y)*x)) + (((r_xy*x)*y)*((x*x)*y)) +
↪  2*(((r_xy*y)*(x*x))*(x*y)) + (((r_xy*y)*x)*((x*x)*y)) + (((r_xy*y)*y)*((x*x)*x)) +
↪  3/2*(((r_yy*x)*(x*x))*(x*y)) - 1/2*(((r_yy*x)*(x*y))*(x*x)) - 2*(((r_yy*x)*x)*((x*y)*x)) +
↪  1/4*(((r_yy*y)*(x*x))*(x*x)) - 1/4*(((x*x)*r_xx)*((y*y)*y)) - 1/2*(((x*x)*y)*((y*y)*r_xx)) -
↪  (((x*y)*(x*y))*(r_xx*y)) - (((x*y)*(y*y))*(r_xx*x)) - 2*(((x*y)*r_xx)*((x*y)*y)) -
↪  (((x*y)*r_xx)*((y*y)*x)) - 2*(((x*y)*r_xy)*((x*y)*x)) - (((x*y)*x)*((y*y)*r_xx))
```

s2_00012_w7

```
degree = 2
weight = 7
d(s2_00012_w7) = -1/2*(((((x*y)*y)*x)*x)*r_yy) - (((((x*y)*y)*x)*y)*r_xy) - 2*(((((x*y)*y)*y)*r_xy)*x) +
↪  (((((x*y)*y)*y)*x)*r_xy) + 1/2*(((((x*y)*y)*y)*y)*r_xx) - 1/4*((((x*x)*y)*(x*y))*r_yy) +
↪  1/4*((((x*x)*y)*r_yy)*(x*y)) - 1/2*((((x*x)*y)*y)*(r_xy*y)) - ((((x*y)*r_xy)*y)*(x*y)) +
↪  ((((x*y)*y)*(x*y))*r_xy) + ((((x*y)*y)*x)*(r_xy*y)) + 1/2*((((x*y)*y)*x)*(r_yy*x)) +
↪  ((((x*y)*y)*y)*(r_xy*x))
boundary_remainder_mod_B_old = 1/4*(((r_xx*x)*(y*y))*(y*y)) + (((r_xx*y)*(y*y))*(x*y)) -
↪  (((r_xx*y)*y)*((x*y)*y)) + 2*(((r_xy*x)*(y*y))*(x*y)) + (((r_xy*x)*x)*((y*y)*y)) +
↪  (((r_xy*x)*y)*((y*y)*x)) + 1/2*(((r_xy*y)*(x*x))*(y*y)) + 2*(((r_xy*y)*(x*y))*(x*y)) +
↪  1/2*(((r_xy*y)*(y*y))*(x*x)) + (((r_xy*y)*x)*((y*y)*x)) + (((r_xy*y)*y)*((x*x)*y)) +
↪  1/4*(((r_yy*x)*(x*x))*(y*y)) + (((r_yy*x)*(x*y))*(x*y)) + 1/4*(((r_yy*x)*(y*y))*(x*x)) -
↪  (((r_yy*x)*x)*((x*y)*y)) - (((r_yy*x)*y)*((x*y)*x)) + (((r_yy*y)*(x*x))*(x*y)) - (((r_yy*y)*x)*((x*y)*x))
↪  - (((x*y)*(y*y))*(r_xx*y)) - (((x*y)*r_xx)*((y*y)*y)) - 2*(((x*y)*r_xy)*((x*y)*y)) -
↪  (((x*y)*r_xy)*((y*y)*x)) - 2*(((x*y)*x)*((y*y)*r_xy)) - (((x*y)*y)*((y*y)*r_xx)) -
↪  1/4*(((y*y)*(y*y))*(r_xx*x)) - 1/2*(((y*y)*r_xx)*((y*y)*x))
selection_remainder_mod_B_old_plus_accepted_reps = 1/4*(((r_xx*x)*(y*y))*(y*y)) + (((r_xx*y)*(y*y))*(x*y)) -
↪  (((r_xx*y)*y)*((x*y)*y)) + 2*(((r_xy*x)*(y*y))*(x*y)) + (((r_xy*x)*x)*((y*y)*y)) +
↪  (((r_xy*x)*y)*((y*y)*x)) + 1/2*(((r_xy*y)*(x*x))*(y*y)) + 2*(((r_xy*y)*(x*y))*(x*y)) +
↪  1/2*(((r_xy*y)*(y*y))*(x*x)) + (((r_xy*y)*x)*((y*y)*x)) + (((r_xy*y)*y)*((x*x)*y)) +
↪  1/4*(((r_yy*x)*(x*x))*(y*y)) + (((r_yy*x)*(x*y))*(x*y)) + 1/4*(((r_yy*x)*(y*y))*(x*x)) -
↪  (((r_yy*x)*x)*((x*y)*y)) - (((r_yy*x)*y)*((x*y)*x)) + (((r_yy*y)*(x*x))*(x*y)) - (((r_yy*y)*x)*((x*y)*x))
↪  - (((x*y)*(y*y))*(r_xx*y)) - (((x*y)*r_xx)*((y*y)*y)) - 2*(((x*y)*r_xy)*((x*y)*y)) -
↪  (((x*y)*r_xy)*((y*y)*x)) - 2*(((x*y)*x)*((y*y)*r_xy)) - (((x*y)*y)*((y*y)*r_xx)) -
↪  1/4*(((y*y)*(y*y))*(r_xx*x)) - 1/2*(((y*y)*r_xx)*((y*y)*x))
```

s2_00013_w7

```
degree = 2
weight = 7
d(s2_00013_w7) = -2*(((((x*y)*y)*y)*y)*r_yy) - (((((y*y)*x)*y)*y)*r_yy) + 2*(((((y*y)*y)*y)*r_yy)*x) -
↪  2*(((((y*y)*y)*y)*x)*r_yy) - 4*(((((y*y)*y)*y)*y)*r_xy) + 6*((((x*y)*y)*y)*(r_yy*y)) +
↪  ((((y*y)*x)*y)*(r_yy*y)) - ((((y*y)*y)*(x*y))*r_yy) + ((((y*y)*y)*r_yy)*(x*y))
boundary_remainder_mod_B_old = -(((r_xy*y)*(y*y))*(y*y)) - 1/2*(((r_yy*x)*(y*y))*(y*y)) -
↪  (((r_yy*y)*(x*y))*(y*y)) - (((r_yy*y)*(y*y))*(x*y)) + (((y*y)*(y*y))*(r_xy*y)) +
↪  2*(((y*y)*r_xy)*((y*y)*y)) + 1/2*(((y*y)*r_yy)*((y*y)*x))
selection_remainder_mod_B_old_plus_accepted_reps = -(((r_xy*y)*(y*y))*(y*y)) - 1/2*(((r_yy*x)*(y*y))*(y*y)) -
↪  (((r_yy*y)*(x*y))*(y*y)) - (((r_yy*y)*(y*y))*(x*y)) + (((y*y)*(y*y))*(r_xy*y)) +
↪  2*(((y*y)*r_xy)*((y*y)*y)) + 1/2*(((y*y)*r_yy)*((y*y)*x))
```

s2_00014_w7

```
degree = 2
weight = 7
d(s2_00014_w7) = -(((((y*y)*y)*y)*y)*r_yy) + ((((y*y)*y)*y)*(r_yy*y))
boundary_remainder_mod_B_old = -1/2*(((r_yy*y)*(y*y))*(y*y)) + 1/2*(((y*y)*r_yy)*((y*y)*y))
selection_remainder_mod_B_old_plus_accepted_reps = -1/2*(((r_yy*y)*(y*y))*(y*y)) +
↪  1/2*(((y*y)*r_yy)*((y*y)*y))
```

s2_00015_w8

```
degree = 2
weight = 8
d(s2_00015_w8) = -((((((x*x)*x)*x)*x)*x)*r_xx) + 1/2*(((((x*x)*(x*x))*x)*x)*r_xx) -
↪  1/2*(((((x*x)*x)*x)*r_xx)*(x*x)) + (((((x*x)*x)*x)*x)*(r_xx*x))
boundary_remainder_mod_B_old = 1/4*((((x*x)*x)*((x*x)*x))*r_xx) - 1/4*((((x*x)*x)*(r_xx*x))*(x*x))
selection_remainder_mod_B_old_plus_accepted_reps = 1/4*((((x*x)*x)*((x*x)*x))*r_xx) -
↪  1/4*((((x*x)*x)*(r_xx*x))*(x*x))
```

s2_00016_w8

```
degree = 2
weight = 8
d(s2_00016_w8) = -((((((y*y)*y)*y)*y)*y)*r_yy) + 1/2*(((((y*y)*(y*y))*y)*y)*r_yy) -
↪  1/2*(((((y*y)*y)*y)*r_yy)*(y*y)) + (((((y*y)*y)*y)*y)*(r_yy*y))
```

```
boundary_remainder_mod_B_old = 1/4*((((y*y)*y)*((y*y)*y))*r_yy) - 1/4*((((y*y)*y)*(r_yy*y))*(y*y))
selection_remainder_mod_B_old_plus_accepted_reps = 1/4*((((y*y)*y)*((y*y)*y))*r_yy) -
↪  1/4*((((y*y)*y)*(r_yy*y))*(y*y))
```

Validation Summary

```
all_completed_weights_passed: True
all_reported_classes_independent_mod_B_old: True
all_reported_z_are_cycles: True
all_reported_z_have_nonzero_B_old_remainder: True
applies_Q_is_false: True
constructs_higher_cells_V3_plus_is_false: True
d_squared_zero_on_formal_s_cells: True
exact_rational_arithmetic: True
report_language_has_no_forbidden_global_phrasing: True
sanity_exp009_low_weight_agreement: {'status': 'checked', 'interpretation': 'Compatibility check only; EXP011
↪  does not use EXP009 as input or as mathematical authority.', 'exp009_selected_weight_bound': 8,
↪  'exp009_degree_2_cell_count': 16, 'exp011_cumulative_v2_cell_count': 16, 'agrees': True}
```